



SUMMARY OF PRODUCT CHARACTERISTICS

M-KAST
Montelukast Chewable Tablets
Rx Only

NAME OF DRUG PRODUCT: Montelukast Chewable Tablets 4 mg
Montelukast Chewable Tablets 5 mg

(TRADE) NAME OF PRODUCT: M-KAST 4
M-KAST 5

STRENGTHS: 4 mg & 5 mg

PHARMACEUTICAL DOSAGE FORM: Chewable Tablet

QUALITATIVE AND QUANTITATIVE COMPOSITION:
M-KAST 4: Each chewable tablet contains Montelukast Sodium Ph.Eur. equivalent to Montelukast 4 mg.
M-KAST 5: Each chewable tablet contains Montelukast Sodium Ph.Eur. equivalent to Montelukast 5 mg.

PHARMACEUTICAL FORM:
M-KAST 4: Pink coloured, mottled, oval, biconvex, uncoated tablets, debossed with 'X' on one side and '52' on the other side.
M-KAST 5: Pink coloured, mottled, round, biconvex, uncoated tablets, debossed with 'X' on one side and '53' on other side.

CLINICAL PARTICULARS:

Therapeutic indications:
Montelukast is indicated for the prophylaxis and chronic treatment of asthma in adults and pediatric patients 12 months of age and older.

Montelukast is indicated in adults and pediatric patients 2 years of age and older for the relief of daytime and nighttime symptoms of seasonal allergic rhinitis.

Posology and method of administration:
Montelukast should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs.

Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

Pediatric Patients 2 to 5 Years of Age with Asthma and/or Seasonal Allergic Rhinitis

The dosage for pediatric patients 2 to 5 years of age is one 4mg chewable tablet daily.

Pediatric Patients 6 to 14 years of Age with Asthma and/or Seasonal Allergic Rhinitis

The dosage for pediatric patients 6 to 14 years of age is one 5mg chewable tablet daily.

General Recommendations

The therapeutic effect of Montelukast on parameters of asthma control occurs within one day. Montelukast chewable tablets can be taken with or without food. Patients should be advised to continue taking Montelukast while their asthma is controlled, as well as during periods of worsening asthma.

No dosage adjustment is necessary for pediatric patients, for the elderly, for patients with renal insufficiency, or mild-to-moderate hepatic impairment, or for patients of either gender.

Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent. Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.

Therapy with Montelukast in Relation to Other Treatments for Asthma

Montelukast can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy:

Bronchodilator Treatments: Montelukast can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled Corticosteroids: Treatment with Montelukast provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. Montelukast should not be abruptly substituted for inhaled corticosteroids.

Contraindications:

Hypersensitivity to the active substance or to any of the excipients

Special Warnings and Precautions for use:

Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled β -agonist should be used. Patients should seek their doctor's advice as soon as possible if they need more inhalations of short-acting β -agonists than usual.

Montelukast should not be abruptly substituted for inhaled or oral corticosteroids. There are no data demonstrating that oral corticosteroids can be reduced when montelukast is given concomitantly.

In rare cases, patients on therapy with anti-asthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.

Montelukast contains aspartame. Contains a source of phenylalanine. May be harmful for people with phenylketonuria.

WARNING

Unsuitable for phenylketonurics.

Effects on ability to drive and use machines

Montelukast has no or negligible influence on the ability to drive and use machines. However, drowsiness or dizziness may occur.

Interaction with other medicinal products and other forms of interaction:

Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following medicinal products: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl oestradiol/norethindrone 35/1), terfenadine, digoxin and warfarin.

The area under the plasma concentration curve (AUC) for montelukast was decreased approximately 40% when co-administration with phenobarbital. Since montelukast is metabolised by CYP 3A4, 2C8, and 2C9, caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, 2C8, and 2C9, such as phenytoin, phenobarbital and rifampicin.

Montelukast is not anticipated to markedly alter the metabolism of medicinal products metabolised by this enzyme (eg., paclitaxel, rosiglitazone, and repaglinide).

Pregnancy and Lactation

Use during pregnancy

Animal studies do not indicate harmful effects with respect to effects on pregnancy or embryonal/foetal development.

Limited data from available pregnancy databases do not suggest a causal relationship between Montelukast and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience.

Montelukast may be used during pregnancy only if it is considered to be clearly essential.

Use during lactation

Studies in rats have shown that montelukast is excreted in milk. It is not known if montelukast is excreted in human milk.

Montelukast may be used in breast-feeding mothers only if it is considered to be clearly essential.

Effects on ability to drive and use machines:

Montelukast is not expected to affect a patient's ability to drive a car or operate machinery. However, in very rare cases, individuals have reported drowsiness or dizziness.

Undesirable effects:

Adverse reactions are listed, by System Organ Class and specific Adverse Experience Term, in the table below

System organ class	Adverse experience term	Frequency category
Infections and infestations	upper respiratory infection	Very Common
Blood and lymphatic system disorders	increased bleeding tendency	Rare
Immune system disorder	hypersensitivity reactions including anaphylaxis	Uncommon
	hepatic eosinophilic infiltration	Very Rare

■ Black A/s: 165 x 240 mm

 AUROBINDO Packaging Development		Product Name	Component	Item Code	Date & Time
		M-KAST	Leaflet	P1521351	03.05.2019 & 5.20 PM
Team Leader Lovaraju		Customer / Country	Version No.	Reason Of Issue	Reviewed / Approved by
		Malaysia_Unit-7	01	NEW	
Initiator Bharath		Dimensions	No. of Colours : 01		
Artist: 		165 x 240 mm	 21351		
		Pharmacode			
		21351			
Additional Information : WOS-19-000366					Sign / Date

P1521351



System organ class	Adverse experience term	Frequency category
Psychiatric disorders	dream abnormalities including nightmares, insomnia, somnambulism, irritability, anxiety, restlessness, agitation including aggressive behaviour or hostility, depression, psychomotor hyperactivity (including irritability, restlessness, tremor)	Uncommon
	disturbance in attention, memory impairment	Rare
	hallucinations, disorientation, suicidal thinking and behaviour (suicidality)	Very Rare
Nervous system disorder	dizziness, drowsiness paraesthesia/ hypoesthesia, seizure	Uncommon
Cardiac disorders	palpitations	Rare
	epistaxis	Uncommon
Respiratory, thoracic and mediastinal disorders	Churg-Strauss Syndrome (CSS), pulmonary eosinophilia	Very Rare
Gastrointestinal disorders	diarrhoea, nausea, vomiting	Common
	dry mouth, dyspepsia	Uncommon
Hepatobiliary disorders	elevated levels of serum transaminases (ALT, AST)	Common
	hepatitis (including cholestatic, hepatocellular, and mixed-pattern liver injury).	Very Rare
Skin and subcutaneous tissue disorders	rash	Common
	bruising, urticaria, pruritus	Uncommon
	angiooedema	Rare
	erythema nodosum, erythema multiforme	Very rare
Musculoskeletal, connective tissue and bone disorders	arthralgia, myalgia including muscle cramps	Uncommon
General disorders and administration site conditions	pyrexia	Common
	asthenia/fatigue, malaise, oedema,	Uncommon

Postmarketing Experience:
Blood and lymphatic system disorders: thrombocytopenia

Psychiatric disorders:
Obsessive-compulsive symptoms

Overdosage:
Overdose symptoms include reports in adults and children with a dose as high as 1,000 mg (approximately 61 mg/kg in a 42 month old child).

Symptoms of overdose
The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

Management of overdose
No specific information is available on the treatment of overdose with montelukast. It is not known whether montelukast is dialysable by peritoneal- or haemo-dialysis.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmaco therapeutic group: Leukotriene receptor antagonists.
ATC-Code: R03D C03

Mechanism of action
The cysteinyl leukotrienes (LTC4, LTD4, LTE4) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene receptors (CysLT) found in the human airway and cause airway actions, including bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment.

Pharmacodynamic effects
Montelukast is an orally active compound which binds with high affinity and selectivity to the CysLT1 receptor.

Pharmacokinetic properties:

Absorption
Montelukast is rapidly absorbed following oral administration. The mean oral bioavailability is 64%. The oral bioavailability and Cmax are not influenced by a standard meal.
For the 5 mg chewable tablet, the Cmax is achieved in 2 hours after administration in adults in the fasted state. The mean oral bioavailability is 73% and is decreased to 63% by a standard meal.
After administration of the 4 mg chewable tablet to paediatric patients 2 to 5 years of age in the fasted state, Cmax is achieved 2 hours after administration. The mean Cmax is 66% higher while mean Cmin is lower than in adults receiving a 10 mg tablet.

Distribution
Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 litres. Montelukast has minimal distribution across the blood-brain barrier. The concentrations of montelukast at 24 hours post-dose were minimal in all other tissues.

Biotransformation
Montelukast is extensively metabolised. Plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children.
Cytochrome P450 2C8 is the major enzyme in the metabolism of montelukast. Additionally CYP 3A4 and 2C9 may have a minor contribution. Therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6. The contribution of metabolites to the therapeutic effect of montelukast is minimal.

Elimination
The plasma clearance of montelukast averages 45 ml/min. 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. This indicates that montelukast and its metabolites are excreted almost exclusively via the bile.

Characteristics in Patients
No dosage adjustment is necessary for the elderly or mild to moderate hepatic insufficiency. Because montelukast and its metabolites are eliminated by the biliary route, no dose adjustment is anticipated to be necessary in patients with renal impairment. There are no data on the pharmacokinetics of montelukast in patients with severe hepatic insufficiency (Child-Pugh score >9).

PHARMACEUTICAL PARTICULARS

List of excipients
Mannitol, Cellulose, microcrystalline, Hydroxypropyl cellulose, Croscarmellose sodium, Ferric oxide red, Aspartame, Artificial cherry flavor, Magnesium Stearate.

Incompatibilities
None known.

Shelf life
Please refer outer package for expiry date.

Special precautions for storage
Store below 30°C. Protect from light and moisture.

Nature and contents of container
Blister pack: 3 blisters of 10 tablets.

Manufactured by


AUROBINDO
AUROBINDO PHARMA LIMITED,
Unit VII, SEZ, TSIIIC, Plot. No. S1.
Survey No's: 411/P, 425/P, 434/P, 435/P & 458/P,
Green Industrial Park, Polepally Village,
Jedcherla Mandal, Mahaboobnagar District,
Telangana state, India.

Product registration Holder in Malaysia:
Healol Pharmaceuticals Sdn. Bhd.
74-3, Jalan Wangsa Delima 6,
KLSC Wangsa Maju, 53300,
Kuala Lumpur.

DATE OF REVISION OF THIS LEAFLET
May 2019.